PeatLiDAR

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Intro - Client

- Nicholas Pontone
  - PhD candidate in the Department of Geography and Environmental Studies at Carleton University
  - Has done work on classifying peatlands in Canada's boreal forest
  - Communicated through email and in person
- His previous work with Peat LiDAR
  - A Hierarchical, Multi-sensor Framework for Peatland Sub-class and Vegetation Mapping Throughout the Canadian Boreal Forest
  - Gave us a polygon of the area

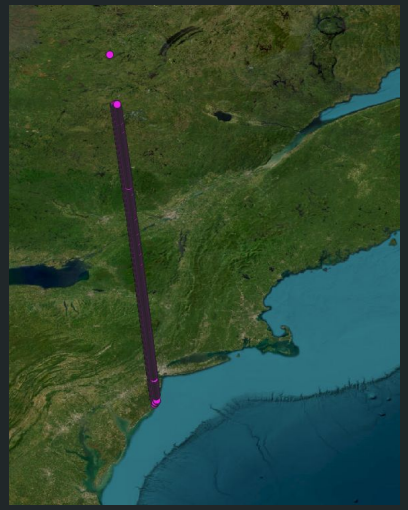
A Hierarchical, Multi-sensor Framework for Peatland Sub-class and Vegetation Mapping Throughout the Canadian Boreal Forest

ORGANIZATION: Carleton University, Canadian Forest Service (CFS), Natural Resources Canada (NRCan)
RESOURCE TYPE: Peer reviewed article

AUTHOR(S): **Nicholas Pontone, Koreen Millard, Dan Thompson, Luc Guindon, André Beaudoin**
ORIGINAL AUTHORS: **Nicholas Pontone, Koreen Millard, Dan Thompson, Luc Guindon, André Beaudoin**
CONTACTS: Nicholas Pontone
RESOURCE DATE: **2024**

Intro - Purpose and Client Requirements

- Improving an existing method for peatland classification
- Goals from Nick
  - Create a tool that:
    - Finds GEDI and ICESat-2 returns that intersect peatlands
    - Put in a manageable file format that can be queried by peatland
    - Aggregate GEDI and ICESat-2 returns by peatland
    - Compare GEDI and ICESat-2 vegetation heights within the peatlands
- Add other satellite information if all goes well



Intro - Data

- Nick shared his Canada peatland polygons shapefile

- GEDI (Global Ecosystem Dynamics Investigation)
  - Full-waveform lidar
  - Designed for vegetation height
  - Attached to the ISS
  - 25m footprint
  - Coverage 51.6˚ N to 51.6˚ S

- ICESat ATL08
  - Photon-counting lidar
  - Apt (automatic picture transmission) for mapping ice, ocean, land elevation
  - Near-polar orbit
  - 16m footprint
  - Coverage: 88˚ N to 88˚ S

Workflow - Config

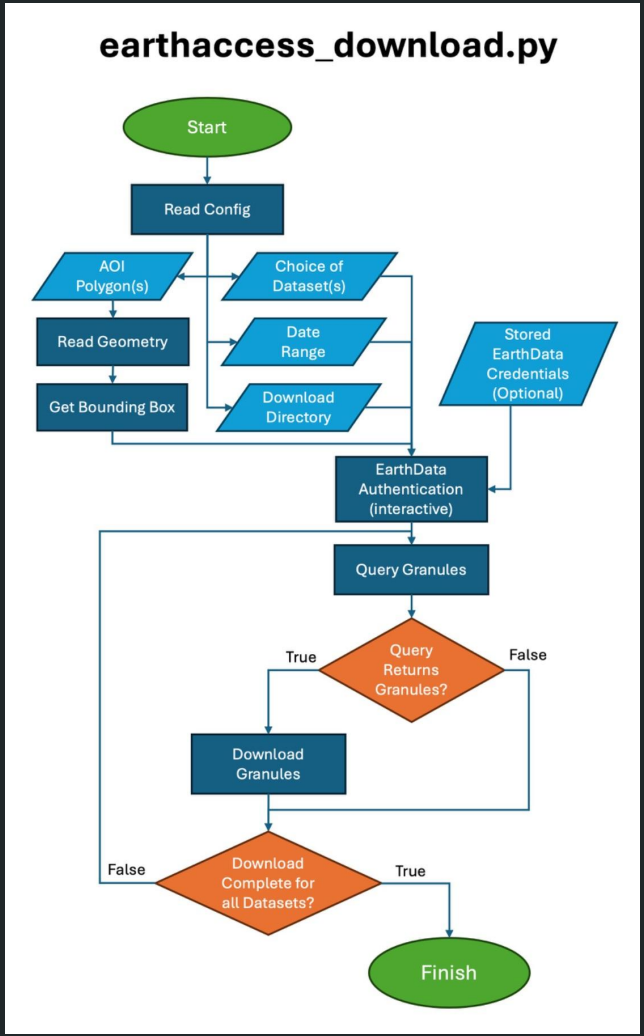
- Configuration file
  - Optional interactive config creation script

- Used for both main scripts
  - earthaccess_download.py
  - process_sat_lidar.py

```
[INPUT]

;Path to the input polygon(s) geometry file (common geospatial formats)
PolygonsPath = Alfred_Bog.kmz

;Path to the directory where data granules will be downloaded
DownloadDir = download_dir

;Short name of the Earthdata product to download (ATL08, GEDI02_A, both)
ShortName = both

;Date range for the earthdata query
StartDate = 2019-01-01
EndDate = 2021-01-01


[ATL08]

;Beam type filter (strong, weak, all)
BeamType = strong

;Night flag filter
NightOnly = True


[GEDI02_A]
QualityFlag = True


[OUTPUT]

;Path to the directory where aggregated files will be written
OutputDir = output

;Output file format options
OutputCSV = True
OutputParquet = True
OutputShapefile = True
```

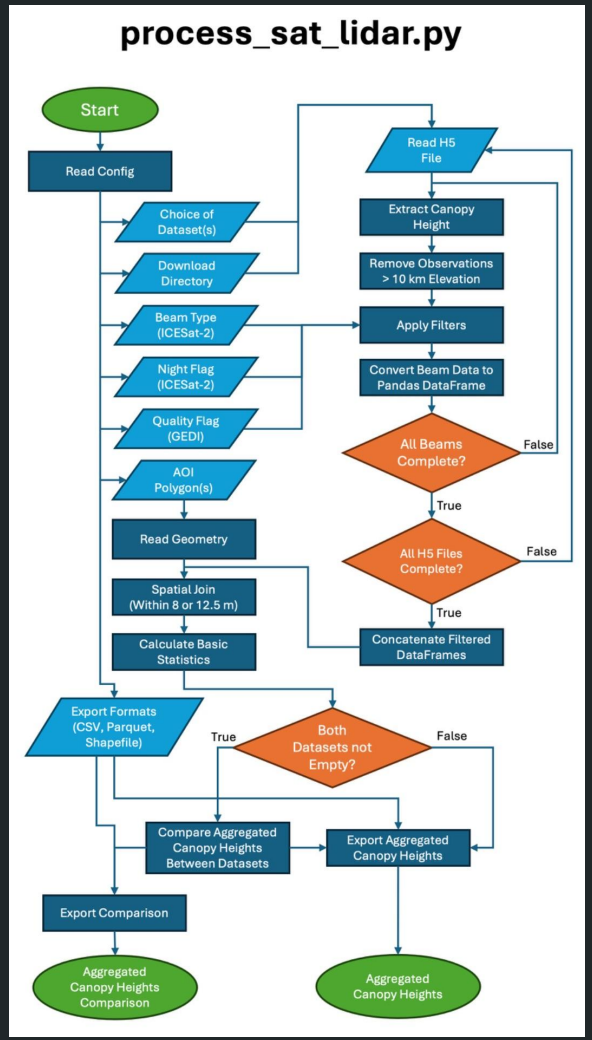
Workflow - Download

- Downloads ICESat-2 & GEDI granules
- NASA EarthData account required (free)
- Uses **earthaccess** Python package
- Avoids duplicate downloads



Workflow - Processing

- Converts HDF5 to Pandas DataFrames
- Various filters applied
- Aggregate by polygon (basic stats)
- Compare between satellites
- Export
  - CSV, Parquet, Shapefile



Setup & Documentation

- Conda environment provided on Github
  - Major Dependencies (Python)
    - earthaccess
    - GeoPandas
    - h5py
    - Configparser

- NASA EarthData account required

- Documentation generated with Sphinx

- scripts package
  - Submodules
  - scripts.earthaccess_download module
    - `authenticate()`
    - `download()`
    - `main()`
    - `search()`
  - scripts.process_sat_lidar module
    - `aggregate()`
    - `compare_data()`
    - `convert_gedi()`
    - `convert_icesat()`
    - `download_check()`
    - `export()`
    - `export_ATL08()`
    - `export_GEDI02_A()`
    - `main()`
    - `read_config()`
  - scripts.user_input3 module
    - `write_user_config()`

Demo

- 268 Sample Polygons - Alfred Bog (on GitHub)
    - Download will not be shown

Challenges

- Conceptualizing satellite lidar data
- Compatibility issues when linking functions
- Slow testing with large data files
- GeoPandas version issues

Limitations

- Requires large downloads
  - Most data is discarded
- Not tested on full date range
- Authentication can be slow
  - Credentials can be stored locally

Future Work

- Investigate other cloud processing options
  - Or ways to limit unnecessary downloads
- Parallel Processing
- Integrate additional data
  - Canopy height percentiles
  - other EarthData products
- Implement accurate ICESat segments in spatial join

Conclusion

- Successful in creating the tool
- Some issues along the way
- Cloud processing would improve workflow
- Good framework for future additions

Acknowledgements

- Nick Pontone
  - Shared example code for ICESat-2 parsing
  - Previous work on Peatlands
  - Data given